

RAHUL MITRA

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EDUCATION

National University of Singapore

Aug 2023 – Jul 2027

Bachelor of Engineering, Industrial Systems Engineering

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Operations Research (NUS System Design Project), Contract, STMicroelectronics

Aug 2026 – Present

- Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.

Robotics Vision Engineer, Amazon (BlendED AI+X)

Jul 2026 – Present

- Build vision systems for 3 imaging workstreams: camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 – Present

- Design a tool loop where a model emits schema-constrained JSON and the solver computes every number, gated by an automated numeric-fidelity check and a per-call audit log with no write path to plant control.

Data Analytics & Product Management Intern, Abbott Laboratories

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Productionized a team-built APC simulator for live internal users on Azure App Service with Docker packaging, access restrictions, and runtime and session hardening.

PROJECTS

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams)

Jan 2026

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.

Competitor, DSTA BRINHACK 2025 CTF

Feb 2025

- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, finishing around 90th of 400+ teams.

LEADERSHIP AND ACTIVITIES

YesWeHack, Independent Bug Bounty Researcher

May 2024 – Present

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and access-control weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports with Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.
- Applied Burp Suite for MitM HTTP interception and Wireshark for OSI Layer 3-4 deep packet inspection and TLS handshake analysis.

NTUC Health Active Ageing Centre, Elderly Care Volunteer

Sep 2025 – Sep 2026

- Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.

Republic of Singapore Navy, Base Support Assistant

Jan 2022 – Jan 2023

- Compressed a 6-month food-waste analysis cycle into a 3-minute reporting process with an Excel VBA automation workflow inside a security-constrained environment.

SKILLS AND CERTIFICATIONS

Technical: Burp Suite, Wireshark, SSRF, CSRF, SQL/NoSQL injection, authentication bypass, OAuth2/JWT, OWASP Top 10, fuzzing, reverse engineering, custom exploit scripting, Python, Bash, Rust, Claude Code, ChatGPT, Gemini, ComfyUI
Certifications: Reverse Engineering Windows Executables (digital forensics); Docker for the Absolute Beginner (Stone River eLearning)